

SOFT2412/COMP9412 Assignment Agile Software Development Practices

Unit of Study: SOFT2412/COMP9412

Sprint Number: Sprint 2

Tutorial time: Friday 2pm-4pm

Tutor name: Amyra Meidiana

DECLARATION

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Project Team Members		
Student Name	Student ID	Signature
Zhiwei Cheng	540249451	Zhiwei Cheng
Xuejian Fang	530313995	Xuejian Fang
Boning Zhao	530027564	Boning Zhao
Xuanhao Yu	530349673	Xuanhao Yu

1. Project Overview

1.1 Project Goals and Objectives

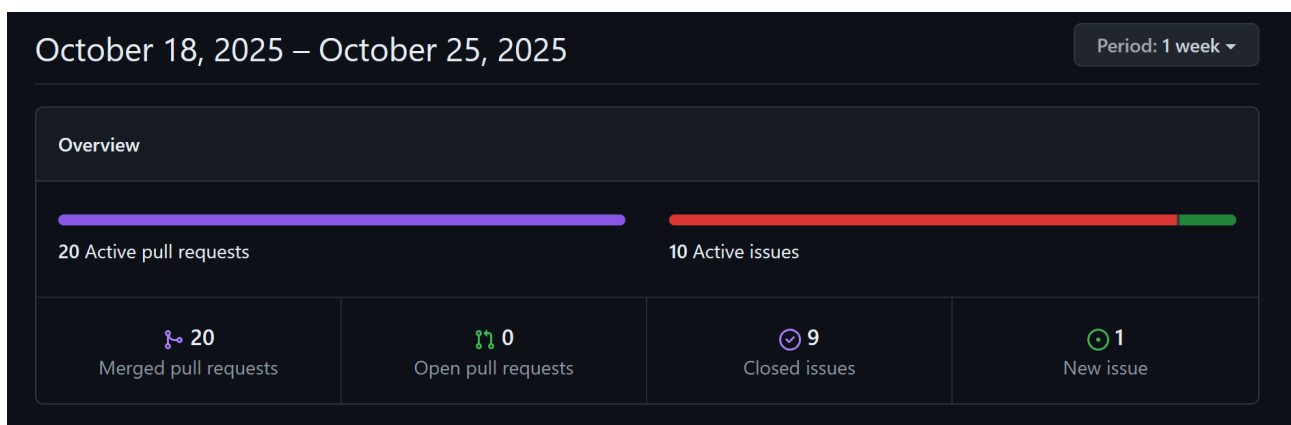
Project goal

Build the Virtual Scroll Access System (VSAS): a Java (Java-17) application that lets users register/login, browse and manage “digital scrolls” (binary files). Guests browse read-only, authenticated users to upload and manage “scrolls”, and administrators to manage users and review usage. The project follows Scrum with time-boxed sprints and CI support. Sprint 2 focused on turning the CLI into a reliable daily tool by completing the scroll lifecycle, reinforcing account security, and hardening release automation so we can demonstrate stable increments to the client.

Objectives set for the project

- Implemented authenticated profile maintenance so users can keep their details current, including secure password rotation and session invalidation, satisfying Assignment 3.1 "Update User Profiles".
- Completed uploader-driven lifecycle controls: update metadata/files and delete obsolete scrolls while enforcing ownership, fulfilling Assignment 3.2 "Scroll Management".
- Extended discoverability and access tooling: download command with safety prompts, scroll preview and search filters let users retrieve information without leaving the CLI.
- Added instrumentation to persist upload/download counters, supporting forthcoming admin reporting.
- Hardened secure credential flows through the shared masked password prompt, aligning with the security expectations in the brief.

Sprint-2 evidence of features implemented and objectives met



Update profile: [PR#84](#)

Edit own scroll: [PR#78](#)

Remove own scroll: [PR#73](#)

Download from list: [PR#72](#)

Search & filters: [PR#80](#), [PR#86](#)

Preview scrolls: [PR#81](#)

Track upload/download counts: [PR#82](#)

Secure password input (masked prompt): [PR#79](#), [PR#85](#),

1.2 Sprint Goals

Sprint 2 targeted three board-level goals aligned with the assignment objectives:

- Deliver complete scroll lifecycle management for authenticated uploaders so the tutor can review end-to-end artefacts.
US-C2 and **US-C3** closed via **PRs #78** and **#73** with version tags **V2.3.0** and **V2.1.0**.
- Improve user trust in the CLI by securing credentials and providing safer data access paths.
US-G1 (PRs #79/#85), **US-D2 (#72)**, and **US-D4 (#81)** confirm the security-enhanced commands.
- Keep [release](#) automation production-ready to support client demos, with the release tooling commits (**#60–#62**) shipping tagged builds **V2.0.0 - V2.8.0**.

These goals were mirrored on the sprint board captured in [Github Project](#), where all Sprint 2 backlog items moved to Done before the client demonstration.

Total Sprint-2 issues: 8/8 CLOSED, including **8 user stories CLOSED** and **34 subtasks completed**.

Evidence from the project board (Sprint-1).

The screenshot displays a Jira project board for 'A3-T28-G03'. The left sidebar shows the 'Sprint' dropdown set to 'Sprint 2' (Oct 20 - Oct 26) with 10 items. The main board area shows a list of 10 backlog items, all of which are 'Done'. The items are as follows:

Title	Assignees	Status
1 US-A3 Update profile #64	bzha8676	Done
2 US-C2 Edit own scroll #66	xuyu8020	Done
3 US-D2 Download from list #68	zche0826	Done
4 US-D3 Search & filters #69	xfan0282	Done
5 US-D4 Preview scrolls #63	bzha8676 and zche...	Done
6 US-C3 Remove own scroll #67	xuyu8020	Done
7 US-E1 Track upload/download counts #70	zche0826	Done
8 US-G1 Secure password input (masked prompt) #71	xfan0282	Done
9 US-UX1 in-app help for existing commands #65	bzha8676	In Progress
10 Cl: switch to Coverage plugin and publish JaCoCo HTML #55	xfan0282	Done

☐	☒	US-G1 Secure password input (masked prompt) #71 by bzha8676 was closed yesterday 4 tasks done	1 2	
☐	☒	US-E1 Track upload/download counts #70 by bzha8676 was closed yesterday 4 tasks done	1 1	
☐	☒	US-D3 Search & filters #69 by bzha8676 was closed yesterday 4 tasks done	1 1	
☐	☒	US-D2 Download from list #68 by bzha8676 was closed yesterday 5 tasks done	1 1	
☐	☒	US-C3 Remove own scroll #67 by bzha8676 was closed yesterday 4 tasks done	1 1	
☐	☒	US-C2 Edit own scroll #66 by bzha8676 was closed yesterday 4 tasks done	1 1	
☐	☒	US-A3 Update profile #64 by bzha8676 was closed yesterday 5 tasks done	1 1	
☐	☒	US-D4 Preview scrolls #63 by bzha8676 was closed yesterday 4 tasks done	1 1	

User stories in Sprint-1:

US code	US-A3	US-C2	US-C3	US-D2	US-D3	US-D4	US-E1	US-G1
issue #	#64	#66	#67	#68	#69	#63	#70	#71

1.3 User Stories

User stories considered in Sprint-2.

- **US-A3 Update profile.** Update profile (PR #84) lets authenticated users update email, phone, and passwords with post-change logout. deliverable in this sprint (tasks closed).
- **US-C2 Edit own scroll.** Give uploaders control over their content deliverable in this sprint (tasks closed).
- **US-C3 Remove own scroll.** Give uploaders control over their content deliverable in this sprint (tasks closed)
- **US-D2 Download from list.** Provide efficient retrieval for users deliverable in this sprint (tasks closed).
- **US-D3 Search & filters.** Provide efficient retrieval for users deliverable in this sprint (tasks closed).
- **US-D4 Preview scrolls.** Surfaces metadata and content snippets for quick assessments deliverable in this sprint (tasks closed).
- **US-E1 Track upload/download counts.** Lays the groundwork for admin reporting deliverable in this sprint (tasks closed).
- **US-G1 Secure password input** (masked prompt). Unifies the login/register experience deliverable in this sprint (tasks closed).
- **US-UX1 in-app help for existing commands** planned, not delivered, will be added in the next Sprint

Changes during the sprint.

US-UX1 (in-app help) remains on the backlog; its detailed plan is [#65](#) and will be scheduled for Sprint 3 to respond to tutor feedback about discoverability.

Changes due to requirements (post-demo, scheduled for Sprint-3).

A new stakeholder request emerged during the Sprint 2 demonstration: introduce bookmark functionality so users can flag scrolls they revisit often. We have logged it for prioritisation in

Sprint 3. No other user story changes were required this sprint, and all committed stories met their acceptance criteria.

2. Scrum Artefacts and Methodologies

2.1 Sprint Planning

Sprint Planning was held at the start of Week 11 with all members present.

Sprint Date: Week 11

Sprint Duration: 1 week

Sprint Goal:

The second sprint, conducted during Week 11, aimed to extend the functionality of the *Virtual Scroll Access System (VSAS)* based on feedback from the previous sprint and to further enhance the CI/CD automation pipeline. The sprint planning meeting was held at the start of the week, where the team discussed development priorities, assigned user stories, and refined the existing backlog to ensure continuous and incremental delivery.

Boning Zhao, serving as the *Product Owner*, was responsible for generating new user stories, assigning tasks to team members, creating issues on GitHub, and managing the overall storyboard. In addition to coordination work, Boning implemented the Upload Profile and Preview Scrolls functionalities and performed related testing to verify correctness and integration stability.

Xuejian Fang, acting as the *Scrum Master*, implemented and tested the Search and Filter features and the Secure Password Input (masked prompt) module. He also set up automated build and release processes within the Jenkins pipeline, enabling the system to automatically build the JAR file, generate version numbers, and publish corresponding GitHub releases and tags. In addition, Xuejian configured code review rules that require each pull request to be approved by at least one reviewer before merging. He maintained and deployed the team's Jenkins server and added testing visualisation plugins such as the Coverage plugin for report generation and review.

Xuanhao Yu was responsible for implementing the Edit Own Scroll and Remove Own Scroll functionalities, enabling users to modify or delete scrolls they previously uploaded. He conducted corresponding tests to ensure that only authorised users could perform these operations and that data consistency was maintained across sessions.

Zhiwei Cheng implemented the Track Upload/Download Counts, Download from List, and Preview Scrolls features. These functions collectively provided administrators and users with accurate usage statistics and improved user interaction with the system. Zhiwei also performed integration testing to verify download correctness and ensured coverage metrics met the sprint's testing standards.

2.2 Daily Stand-ups

All stand-up meetings for this sprint were conducted **in person**, allowing the team to communicate more efficiently and collaborate directly on technical issues. The first meeting was held on **Monday**, marking the official start of Sprint 2. During this session, the *Product Owner* reviewed the shortcomings and challenges identified in the previous sprint, particularly regarding usability and user interaction. He then outlined the objectives for the current sprint, assigned new tasks to each team member, and updated the project storyboard accordingly.

The second meeting took place on **Wednesday**, where each member presented their progress and demonstrated the features they had implemented. These included both new functional components and improvements to testing coverage through the updated CI/CD pipeline. The results were largely satisfactory, as most of the planned user stories had been successfully completed and passed verification. However, several areas were identified for further improvement, such as optimising the terminal input mode for better user experience and refining the overall product presentation for future iterations.

Overall, the stand-ups effectively maintained project transparency, aligned the team's goals, and ensured that all members stayed focused on the sprint objectives.

2.3 Retrospective

The sprint retrospective meeting was held at the end of Week 11, with all team members attending. The discussion focused on evaluating the outcomes of Sprint 2, reviewing progress against the sprint goal, and identifying areas for improvement in both technical and procedural aspects.

The team successfully achieved most of the planned objectives for this iteration. Core new features, including search and filtering, profile uploading, scroll editing and deletion, as well as download and preview functionalities, were implemented and tested effectively. The integration of automated versioning and release creation within the Jenkins pipeline greatly improved the project's deployment efficiency. The introduction of mandatory code review rules also enhanced code quality and reduced merge conflicts. In addition, the testing visualisation plugins in Jenkins provided better insight into test coverage and build stability, helping the team monitor CI/CD performance in real time.

Despite these achievements, several aspects were identified for improvement. The command-line input mode, while functional, still lacks user-friendliness and could benefit from further optimisation. The team also discussed enhancing the visual design and user interface presentation in future iterations to improve overall usability. Furthermore, although the automation workflow was successful, a few build delays and dependency issues were observed, indicating the need for further pipeline refinement and configuration tuning.

Moving forward, the team plans to refine the terminal interface for a smoother user experience, begin initial planning for UI visualisation, and improve Jenkins performance by optimising build stages and caching dependencies. The team will also continue enforcing code review standards and maintain a focus on improving test coverage and release stability.

2.4 Use of Scrum Artefacts

All Scrum artefacts for Sprint 2, including the Product Backlog, Sprint Backlog, and User Stories, are maintained on the GitHub Projects board. The board records task priorities, assignments, and progress status throughout the sprint.

Project backlog and Sprint backlog:

<https://github.sydney.edu.au/orgs/SOFT2412-COMP9412-2025s2/projects/309/views/1>

The team also improved the visibility of the CI/CD process through Jenkins. Automated builds, testing, and versioned releases were integrated into the Jenkins pipeline, which generated visual

test and coverage reports through installed plugins. Evidence of these artefacts and automation outputs can be accessed via the team's GitHub repository and Jenkins interface:

The Jenkins Console Output:

https://github.sydney.edu.au/SOFT2412-COMP9412-2025s2/A3-T28-G03/tree/Appendix_A

3. Agile Development Tools and Practices

3.1 Version Control

3.1.1 Tools and Branching Strategy

Unchanged from last week: Git + GitHub (Issues, Projects, PRs), Gradle builds/tests with google-java-format, and the feature-branch model (feature/<story-name>), conventional commit prefixes, and PR-only merges to a protected main.

New this sprint — branch protection rules on main:

- Require a pull request before merging.
- Require status checks to pass (Jenkins pipeline must be green for the PR's head SHA).
- Require approvals: at least 1 reviewer must approve before merge.
- Block force pushes.

These rules formalize our "no direct commits to main" policy and ensure CI + review gates are consistently enforced.

Note: Overall toolchain is unchanged; Jenkins pipeline changes (auto jar build, auto versioning/release) are documented in 3.2 CI/CD Tools and Practices this sprint.

3.1.2 Merge Conflicts and Branch Integration

Process remains as last week (frequent rebases/merges from main, full local test/format before push).

New incident this sprint: one teammate opened a PR without pulling latest main, leading to a PR that included unrelated historical commits. Resolution: we used squash merge to collapse the entire PR into a single cohesive commit after review, preserving a clean history.

3.1.3 How Added Features Fit User Requirements

Following the Week-11 tutorial demo of Sprint 2, the client asked for a more secure way to enter passwords so that they aren't visible while typing. In Sprint 2 we delivered US-G1 Secure password input: when users run register or login without a --password, the CLI now asks for a password directly in the terminal without echoing characters, and prevents mistakes by detecting mismatches where appropriate. This change matches the client's goal—protect credentials during normal use—and keeps the overall registration/login flow unchanged for existing users, satisfying the story's acceptance criteria seen in the demo.

merge pull request #86 from SOFT2412-COMP9412-20252/chore/remove-legacy-list-output Boring Zhao
test(list): update tests to verify fixed-width output format bzhah876

refactor(list): remove legacy output for backwards compatibility bzhah876

merge pull request #85 from SOFT2412-COMP9412-20252/fix/login-single-prompt-and-usage Boring Zhao

test(login): add tests for single password prompt and fast-fail usage when username missing xfan0282

fix(login): prompt for password once; require username before prompting and show usage xfan0282

merge pull request #84 from SOFT2412-COMP9412-20252/feat/a3-a3-update-profile Cheng Zhiwei Chang

merge branch 'feat/a3-a3-update-profile' of https://github.com:edu.cn/soft2412-COMP9412-20252(A3-T28-G03) into feat/a3-a3-update-profile bzhah876

test(repo): add test for handling rows with missing trailing columns in profile update bzhah876

fix(profile): handle missing fields during user profile updates in TSV bzhah876

test(profile): add tests for updating user profile fields and password handling bzhah876

fix(profile): add profile update command and repository method bzhah876

test(repo): add test for handling rows with missing trailing columns in profile update bzhah876

fix(profile): handle missing fields during user profile updates in TSV bzhah876

test(repo): add tests for updating user profile fields and password handling bzhah876

feat(profile): add profile update command and repository method bzhah876

merge pull request #82 from SOFT2412-COMP9412-20252/feat/ut-e1-track-scroll-usage Boring Zhao

fix: correct comment typo in UTF-8 handling logic Zcho0826

feat: track scroll upload/download counts - US-E21 Zcho0826

merge pull request #81 from SOFT2412-COMP9412-20252/feat/ut-d4-preview-sroll Cheng Zhiwei Chang

fix(dispatcher): add preview case Zcho0826

feat(preview): enhance UTF-8 handling and add tests for text and binary previews Zcho0826

refactor(tests): clean up formatting and remove unused imports in PreviewCommandTest Zcho0826

feat(cli): add preview command to display scroll metadata and content snippet Zcho0826

merge pull request #80 from SOFT2412-COMP9412-20252/feat/d4-search-filters Boring Zhao

fix(cli): print fixed-width table before legacy rows to match first 519 file as fixed-width xfan0282

fix(list): also print legacy pipe-separated rows to keep backward-compatible tests green xfan0282

test(list): cover filters, invalid date errors, empty-state message, and fixed-width table stability xfan0282

feat(list): stable fixed-width table output xfan0282

fix(list): make captured filter variables final for stream; remove unused import xfan0282

feat(list): parse filters and validate dates (yyyy-mm-dd) xfan0282

merge pull request #79 from SOFT2412-COMP9412-20252/feat/c1-secure-password-prompt Boring Zhao

test(cli): add tests for PasswordPrompt and interactive password flows xfan0282

fix(register): check non-password required flags before prompting xfan0282

feat(login): support masked password prompt when --password is omitted xfan0282

fix(register): prompt for masked password when --password is omitted xfan0282

feat(cli): add PasswordPrompt with masked input, fallback, and test seam xfan0282

merge pull request #78 from SOFT2412-COMP9412-20252/feat/scroll-detect-detection-anchor Xuejian Fang

fix: fixdate comment Xuejian Fang

fix(release): anchor feat detection to PR source branch and support squash merges xfan0282

merge pull request #76 from SOFT2412-COMP9412-20252/feat/scroll-detect Xuejian Fang

style(cli): apply Google JavaFormat for US-C2 scroll update Xuehiao

test(scroll): add ScrollUpdateSubcommand tests for US-C2 update feature (85% instruction and 76% branch coverage) Xuehiao

feat(scroll): implement scroll update command for US-C2 Xuehiao

merge pull request #75 from SOFT2412-COMP9412-20252/feat/c1-release-feat-detect Xuejian Fang

fix(release): simplify feat regex to avoid Groovy escape error xfan0282

fix(release): escape parentheses in feat regex xfan0282

fix(release): detect feat commits from squash merges xfan0282

merge pull request #73 from SOFT2412-COMP9412-20252/feat/scroll-delete Xuehiao Yu

test(scroll): add ScrollCommand tests to cover CLI usage and subcommands Xuehiao

test(scroll): add ScrollDeleteSubcommand test (84% instruction, 70% branch coverage) Xuehiao

fix(cli): update interactive prompt text and reformat code to Google Java Style Xuehiao

fix(scroll): support 'scroll delete -id <id>' for owner-only deletion Xuehiao

Feat(d2 download scroll) (#72) Cheng Zhiwei Chang

merge pull request #62 from SOFT2412-COMP9412-20252/ci/release-jar Xuejian Fang

feat(cli): build fat jar with shadowJar and upload it to GitHub Release xfan0282

feat(build): add Shadow plugin to produce fatJar with minimal changes xfan0282

merge pull request #61 from SOFT2412-COMP9412-20252/fix/coverage-adaptor-comment Xuejian Fang

fix(cli): drop jacocoAdaptor usage; use legacy parser map for recordCoverage and guard it xfan0282

merge pull request #60 from SOFT2412-COMP9412-20252/fix/release-gh-headsha-idempotent Xuejian Fang

fix(release): set HEAD_SHA on main and make gh release creation idempotent xfan0282

merge pull request #59 from SOFT2412-COMP9412-20252/chore/c1-server-release Xuejian Fang

fix(release): disable test-mode to release runs only on main xfan0282

fix(release): drive gh with GH_HOST/GH_TOKEN (remove --hostname), write CHANGELOG.txt, and guard release idempotently xfan0282

ci(release): switch to GitHub CLI for tagging and release and add release author xfan0282

ci(release): set gh commit identity before tagging (fix test-mode release) xfan0282

ci(release): temporarily enable real tag/release on non-main to validate flow xfan0282

ci(release): auto tag & GitHub release on main (API-first + title fallback) xfan0282

ci: make API retry POSIX-safe and normalize tag prefix (Vw) xfan0282

ci: fix dry-run auth for fetching tags (use HTTPS PAT) xfan0282

ci: add API-first next-version calculator with tag dry-run on non-main) xfan0282

chore(cli): add VERSION_MAJOR xfan0282

merge pull request #58 from SOFT2412-COMP9412-20252/fix/grade-cases-perms Xuejian Fang

fix(cli): correct JReCo HTML path and configure coverage source directories xfan0282

fix(cli): redirect Gradle user home to workspace to avoid EPERM on --gradle; disable implicit checkout xfan0282

merge pull request #56 from SOFT2412-COMP9412-20252/ci/coverage-and-html-publish Boring Zhao

ci: switch to Coverage plugin and publish JaCoCo HTML xfan0282

merge pull request #54 from SOFT2412-COMP9412-20252/test/increase-coverage-rate Xuehiao Yu

test: code style Xuehiao

merge pull request #53 from SOFT2412-COMP9412-20252/test/increase-coverage-rate Xuehiao Yu

add file scroll repository edge test, instruction coverage rate increase to 85% Xuehiao

add file user repository duplicate test, branch coverage rate increase to 60% Xuehiao

add commanddispatchertest, instruction, instruction coverage increase to 84%, branch coverage rate increase to 59% Xuehiao

merge pull request #52 from SOFT2412-COMP9412-20252/fix/cli-usage-help-text Boring Zhao

fix(docs): update test report paths in README bzhah876

delete(cli): remove obsolete scrollsrv file bzhah876

fix(cli): align usage text with commands bzhah876

merge pull request #51 from SOFT2412-COMP9412-20252/chore/c1-gignore-data Xuehiao Yu

add "data" folder in .gitignore Xuehiao

merge pull request #50 from SOFT2412-COMP9412-20252/feat/c1-session Boring Zhao

feat(cli): implement session persistence in Login and Logout commands bzhah876

merge pull request #49 from SOFT2412-COMP9412-20252/feat/require-interactive-shell Boring Zhao

chore: add grade4properties to configure console output bzhah876

Add interactive shell mode for VSAS CLI bzhah876

merge pull request #48 from SOFT2412-COMP9412-20252/feat/ut-a1-register-account Xuejian Fang

test(cli): cover register success/duplicate/missing and no-plaintext (A3-T28-G03#1) xfan0282

feat(cli): enforce unique idKey in register (A3-T28-G03#4) xfan0282

feat(cli): implement usage with salted hashing via repository (A3-T28-G03#3) xfan0282

feat(security): add salted SHA-256 password hasher with tests (A3-T28-G03#1) xfan0282

feat(repo): implement TSV-backed FileRepository with minimal tests (A3-T28-G03#9) xfan0282

feat(model): add User.createdAt and backward-compatible constructor (A3-T28-G03#8) xfan0282

merge pull request #47 from SOFT2412-COMP9412-20252/feat/c1-scroll-list-command Cheng Zhiwei Chang

feat(scroll): add list command with TSV-backed repository Zcho0826

merge pull request #46 from SOFT2412-COMP9412-20252/fix/formatting-register-command-authhashingtest Xuejian Fang

ci(release): add test for authentication for register Command java and AuthhashingIntegrationTest java xfan0282

merge pull request #45 from SOFT2412-COMP9412-20252/feat/ut-a1-hashed-passwords Xuejian Fang

test(auth): add salt-variance test for same password (A3-T28-G03#4) xfan0282

feat(auth): hash password on registration and persist salt (A3-T28-G03#3) xfan0282

feat(auth): verify hashed passwords during login (A3-T28-G03#4) xfan0282

feat(security): add salted password hashing utility (A3-T28-G03#8) xfan0282

merge pull request #44 from SOFT2412-COMP9412-20252/feat/c1-implement-upload-command Cheng Zhiwei Chang

merge branch 'main' into feat/c1-implement-upload-command Xuejian Fang

style(upload): use google-java-format format CopyCommand.java Zcho0826

feat(cli): implement UploadCommand with file copy and metadata persistence Zcho0826

merge pull request #42 from SOFT2412-COMP9412-20252/feat/c1-scroll/c1-upload-unique-id Zhiwei Chang

feat(cli): update CommandDispatcher to return success on usage help and enhance FileRepository and SessionService for scroll management Zcho0826

merge pull request #41 from SOFT2412-COMP9412-20252/feat/ut-sprint-c1-skeleton Boring Zhao

chore(cli): add runnable cli skeleton and package structure bzhah876

merge pull request #40 from SOFT2412-COMP9412-20252/feat/ut-a1-gradlew-and-format-check Xuejian Fang

ci: re-enable googleJavaFormat check for non-blocking behavior xfan0282

fix(cli): make gradlew executable for CI and local usage xfan0282

merge pull request #6 from SOFT2412-COMP9412-202522/chore/configure-project-tooling Xuejian Fang

chore(build): configure initial testing and formatting tools bzhah876

merge branch 'main' of https://github.com:edu.cn/soft2412-COMP9412-20252(C28-Team03-A2A3) bzhah876

merge pull request #4 from SOFT2412-COMP9412-20252/fix/ci-jenkinsfile-fix Xuejian Fang

fix(cli): remove unsupported and/or color options, ensure gradlew exec & skip google-java-format main0282

fix(cli): ensure gradlew executable in CI, skip verifyGoogleJavaFormat, publish .html and archive .Jars; safer deploy xfan0282

merge pull request #3 from SOFT2412-COMP9412-20252/chore/bootstrap-with-gradle-init Boring Zhao

chore: initialize project via Gradle init and add CI bzhah876

Revert "chore: add initial build gradle file for Java application setup" bzhah876

chore: add initial build gradle file for Java application setup bzhah876

merge pull request #2 from SOFT2412-COMP9412-202522/chore/init-and-gilgnore Boring Zhao

docs: add README with run/test instructions and project .gitignore bzhah876

merge pull request #1 from SOFT2412-COMP9412-202522/feat/ut-a1-jenkinsfile Xuejian Fang

ci(jenkins): add Jenkinsfile for Gradle multi-branch pipeline xfan028

main

45 Branches 35 Tags

Go to file

Add file

Code

About



bzza8676 Merge pull request #86 from SOFT2412-COMP9412-2025s2/cho... bf9ba9c · 2 days ago 136 Commits

app	test(list): update tests to verify fixed-width output format	2 days ago
gradle	chore: initialize project via Gradle init and add CI	2 weeks ago
.gitattributes	chore: initialize project via Gradle init and add CI	2 weeks ago
.gitignore	add "data" folder in gitignore	last week
Jenkinsfile	fix: update comment	3 days ago
LICENSE	Initial commit	2 weeks ago
README.md	fix(docs): update test report paths in README	last week
VERSION_MAJOR	chore(ci): add VERSION_MAJOR	5 days ago
gradle.properties	chore: add gradle.properties to configure console output	last week
gradlew	fix(ci): make gradlew executable for CI and local usage	last week
gradlew.bat	chore: initialize project via Gradle init and add CI	2 weeks ago
settings.gradle	chore: initialize project via Gradle init and add CI	2 weeks ago

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CC28-Team03-A2A3 - VSAS

Run

```
./gradlew run
```

Test & Coverage

```
./gradlew test jacocoTestReport
```

```
Reports: app/build/reports/tests/test, app/build/reports/jacoco/test/html
```

Code Style (Google Java Format)

```
Format locally: ./gradlew googleJavaFormat
```

```
Verify in CI (already wired to check): ./gradlew verifyGoogleJavaFormat
```

No description, website, or topics provided.

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Releases 35

[V2.8.2](#) Latest
2 days ago

+ 34 releases

Contributors 4

xfan0282 Xuejian Fang
 bzha8676 Boning Zhao
 xuyu8020 Xuanhao Yu
 zche0826 Cheng Zhiwei Cheng

Languages

Java 100.0%

Clear current search query, filters, and sorts

☐	0 Open ✓ 42 Closed	Author -	Label -	Projects -	Milestones -	Reviews -	Assignee -	Sort -
☐	Chore/remove legacy list output ✓ #86 by bzha8676 was merged 2 days ago • Approved							
☐	Fix/login single prompt and usage ✓ #85 by xfan0282 was merged 2 days ago • Approved					1		
☐	Feat/us a3 update profile ✓ #84 by bzha8676 was merged 2 days ago • Approved					1		
☐	feat: track scroll upload/download counts - US-E1 ✓ #82 by zche0826 was merged 2 days ago • Approved					1		
☐	Feat/us d4 preview scroll ✓ #81 by zche0826 was merged 2 days ago • Approved							1
☐	Feat/d3 search filters ✓ #80 by xfan0282 was merged 3 days ago • Approved					1		
☐	Feat/g1 secure password prompt ✓ #79 by xfan0282 was merged 3 days ago • Approved					1		
☐	US-C2 Feat/scroll update ✓ #78 by xuyu8020 was merged 3 days ago • Approved					1		
☐	Feat/us d4 preview scroll ✗ invalid wontfix #77 by bzha8676 was closed 3 days ago • Review required							
☐	fix(release): anchor feat detection to PR source branch and support s... ✓ #76 by xfan0282 was merged 3 days ago • Approved							1
☐	Fix/release feat detection ✓ #75 by xfan0282 was merged 3 days ago • Review required							
☐	fix(release): detect feat commits from squash merges ✗ invalid #74 by xfan0282 was closed 3 days ago • Review required							
☐	US-C3 Feat/scroll delete ✓ #73 by xuyu8020 was merged 3 days ago • Review required					1		
☐	Feat/us d2 download scroll ✓ #72 by zche0826 was merged 4 days ago • Review required					1		
☐	CI/release jar • #62 by xfan0282 was merged 5 days ago • Review required							
☐	fix(ci): drop jacocoAdapter usage; use legacy parser map for recordCo... • #61 by xfan0282 was merged 5 days ago • Review required							
☐	fix(release): set HEAD_SHA on main and make gh release creation idemp... ✗ #60 by xfan0282 was merged 5 days ago • Review required							
☐	Chore/ci semver release • #59 by xfan0282 was merged 5 days ago • Review required							
☐	Fix/gradle cache perms • #58 by xfan0282 was merged 5 days ago • Review required							
☐	chore: deprecated PR • #57 by xfan0282 was merged last week • Review required							
☐	chore: deprecated PR ✗ #56 by xfan0282 was merged last week • Review required							
☐	CI: switch to Coverage plugin and publish JaCoCo HTML ✓ #55 by xfan0282 was merged last week • Review required							
☐	set code style ✓ #54 by xuyu8020 was merged last week • Review required							
☐	Test/increase coverage rate • #53 by xuyu8020 was merged last week • Review required							
☐	Fix/cli usage help text • #52 by bzha8676 was merged last week • Review required							

Rulesets

New branch ruleset



All



Protect main



Active

4 rules • targeting 1 branch



SOFT12412-COMP9412-2025v2 / A3-T28-G03

Q

Type ↵ to search

Code

Issues

Pull requests

Projects

Wiki

Security

Insights

Settings

General

Access

Collaborators and teams

Team and member roles

Code and automation

Branches

Tags

Rules

Rulesets

Insights

Hooks

Pages

Custom properties

Security

Code security and analysis

Deploy keys

Integrations

GitHub Apps

Email notifications

Autolink references

Custom tabs

Rulesets / Protect main

Ruleset Name

Protect main

Enforcement status

Active

Bypass list

Exempt roles, teams, or apps from this ruleset by adding them to the bypass list

Organization admin Role Always allow

Repository admin Role Allow for pull requests only

Targets

Which branches do you want to make a ruleset for?

Target branches

Branch targeting criteria

main

release/*

Applies to 1 target: main

Rules

Which rules should be applied?

Branch protections

Restrict creations

Restrict updates

Restrict deletions

Require linear history

Require deployments to succeed

Require signed commits

Require a pull request before merging

Required approvals

Dismiss stale pull request approvals when new commits are pushed

Require review from Code Owners

Require approval of the most recent reviewable push

Require conversation resolution before merging

Require status checks to pass

Require branches to be up to date before merging

Status checks that are required

continuous-integration/jenkins/pr-merge

Block force pushes

Require code scanning results

Restrictions

Restrict commit metadata

Restrict branch names

Save changes

Revert changes

GitHub Enterprise Server 3.14.18

The University of Sydney

© 2025 GitHub, Inc. Help Support

A3-T28-G03 #71

US-G1 Secure password input (masked prompt) #71

Closed

bzha8676 opened 4 days ago

Edit title

bzha8676 3 days ago (edited)

Edit

Story

As a User, I want the CLI to prompt for passwords without echo, so that my credentials are not exposed.

Acceptance Criteria

Acceptance Criteria

- Running `register` or `login` without `--password` triggers masked prompts asks for `Password:` and `Confirm password:`;
- Mismatch fails with a clear error.

Tasks

- Introduce a reusable password prompt helper (e.g. `org.soft2412.vsas.cli.PasswordPrompt`) that masks input via `Console.readPassword`, falls back to manual char reading when the console is unavailable, and exposes a test seam for injecting canned responses.
- Refactor `RegisterCommand` so `--password` becomes optional: when absent, invoke the password prompt for `Password:` and `Confirm password:`, abort with exit code `1` and a clear mismatch message when they differ, and keep the existing flag flow intact.
- Refactor `LoginCommand` to reuse the password prompt when `--password` is omitted, requiring confirmation and returning exit code `1` with a descriptive message on mismatch.
- Add unit tests for the prompt helper and the interactive branches of `RegisterCommand` / `LoginCommand`, using injected doubles to assert prompt text, mismatch handling, and exit codes.

Assignees

xfan0282

Labels

Add labels...

Milestone

Add milestone...

Status

Done

Linked pull requests

#79

#85

Repository

A3-T28-G03

Priority

P1

Size

M

Estimate

2

Sprint

Q Sprint 2

Start date

Oct 20, 2025

End date

Oct 26, 2025

Open in new tab

Copy link

Copy link in project

Archive

3.2 CI/CD Tools and Practices

3.2.1 Tooling & Overall Approach

We continue to use Jenkins (Multibranch Pipeline) with Gradle for build/test, google-java-format for style verification, JaCoCo for coverage, and GitHub for status reporting. The overall toolchain is unchanged from last week; this sprint focused on strengthening the pipeline script and release automation.

3.2.2 Pipeline Overview (What runs on every branch/PR)

Our Jenkinsfile defines a single, consistent pipeline that every discovered branch and Pull Request runs:

- Checkout: clean workspace, checkout scm.
- Sanity: quick environment diagnostics (working dir, wrapper presence, Gradle cache path).
- Prepare Gradle cache: create and warm \$GRADLE_USER_HOME to speed up builds.
- Format (verify): run `verifyGoogleJavaFormat`. If mismatched, CI surfaces a clear action ("run `./gradlew googleJavaFormat` locally").
- Build: clean assemble shadowJar to produce both regular JARs and a fat/uber JAR (`app-all.jar`) for demos.
- Test & Coverage: test `jacocoTestReport` with JUnit and JaCoCo XML/HTML; reports are published and archived.
- Archive: keep regular jars and the fat jar as build artifacts.
- Deploy (main only; placeholder): no environment changes yet in this stage; the actual release happens in the next stage.
- Release (main only or test mode): compute version, tag the repo, create a GitHub Release, and upload `app-all.jar`.

3.2.3 Status Visibility in GitHub

Each commit and Pull Request shows the Jenkins check on the right-hand side in GitHub. A green check indicates the pipeline passed; failures block merges until resolved. This continues to provide a clear, enforceable quality gate at the PR level.

3.2.4 Build, Test, and Coverage (strengthened this sprint)

Style gate runs early and fails fast with an explicit hint.

Build always produces a shadow JAR app-all.jar for customer demos and GitHub Releases.

Test & Coverage outputs JUnit XML and JaCoCo XML/HTML. Jenkins publishes HTML coverage and archives raw reports, so results remain auditable for the PR and for the release job that follows.

3.2.5 Automated Versioning & Release

We implemented fully automated Git tagging and GitHub Releases in the pipeline:

Versioning rules (encoded in the Jenkinsfile):

- Major is sprint-based and controlled by a file (VERSION_MAJOR)—current sprint is 2, so releases are V2.x.y.
- Minor increments (and patch resets to 0) if the merged branch (or merge subject) matches feat* / feature*.
- Otherwise we patch++.
- Tag prefix is preserved to match prior history (V vs v) and is auto-detected.

Release conditions:

- Only after a PR is merged into main and CI passes do we create a tag and a GitHub Release; non-main branches still build and test, but do not publish.

Current state:

- 35 total versions to date; latest is V2.8.2.

This automation ensures releases are consistent, reproducible, and traceable to the exact CI run—including coverage evidence.

3.2.6 CD Status

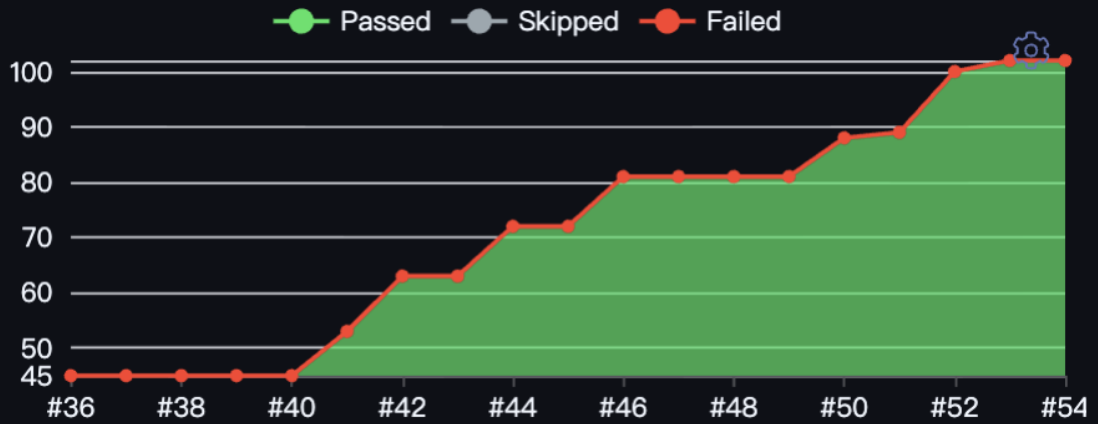
We now perform Continuous Delivery to GitHub Releases: main-branch merges automatically tag, create a release, and upload the runnable artifact. Environment deployment is still out of scope for this sprint; our next step is to wire a test/pre-prod environment that pulls the latest release, runs smoke tests, and supports rollback. Once stable, we will re-evaluate moving towards Continuous Deployment.

3.2.7 Reliability & Hosting (new this sprint)

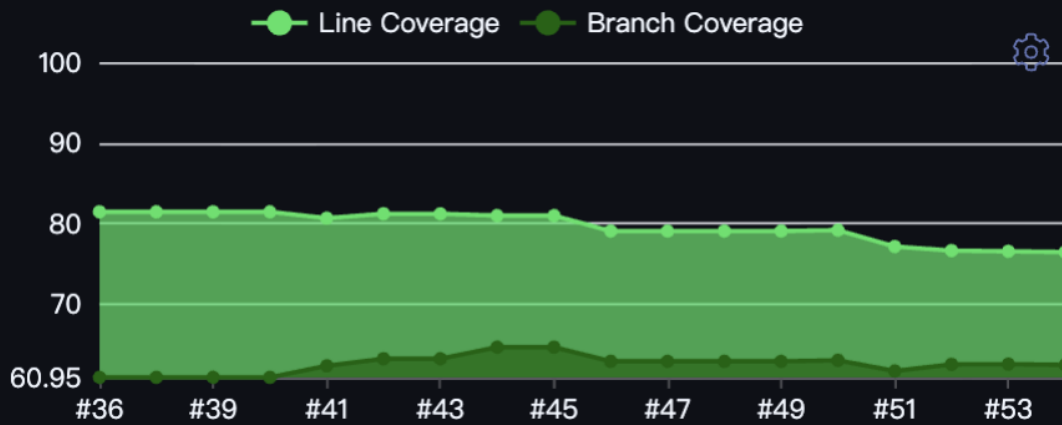
To keep CI available 24x7, we containerised our Jenkins instance and migrated it to a 24/7 server. We also bound the instance to a DNS name for the team's easy access. This reduced intermittent outages and removed "sleeping laptop" risks during review cycles or late-day merges.

Branches (44) Pull Requests (28)

Test Result Trend



Code Coverage Trend



Jenkins / A3-T28-G03 / main

Status

Changes

Build Now

View Configuration

JaCoCo HTML

Favorite

Open Blue Ocean

Stages

GitHub

Coverage Report

Pipeline Syntax

Credentials

main

Full project name: A3-T28-G03/main



Last Successful Artifacts



Latest Test Result (no failures)

Permalinks

- Last build (#54), 2 days 0 hr ago
- Last stable build (#54), 2 days 0 hr ago
- Last successful build (#54), 2 days 0 hr ago
- Last failed build (#47), 2 days 15 hr ago
- Last unsuccessful build (#47), 2 days 15 hr ago
- Last completed build (#54), 2 days 0 hr ago

Builds

Filter

October 24, 2025

#54 4:09 AM

#53 2:49 AM

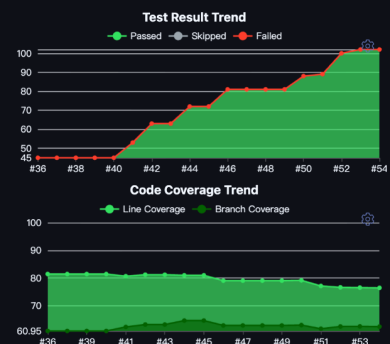
#52 2:06 AM

#51 1:47 AM

#50 12:42 AM

October 23, 2025

#49 2:11 PM



Jenkins

A3-T28-G03

main

#54

Coverage Report

Search

Settings

Menu

User

Status

Changes

Console Output

Edit Build Information

Delete build '#54'

Timings

Git Build Data

Tests

Coverage Report

JaCoCo HTML

See Fingerprints

Open Blue Ocean

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

Previous Build

Coverage Report

OverviewLine CoverageBranch CoverageCyclomatic ComplexityFiles

Coverage of all files

Show only changed files

Search:

File	Package	Line	Branch	LOC	Complexity
LogoutCommand.java	org.soft2412.vsas.cli	60.00%	N/A	10	5
PasswordPrompt.java	org.soft2412.vsas.cli	14.29%	5.00%	35	14
PreviewCommand.java	org.soft2412.vsas.cli	80.00%	63.51%	115	46
LoginCommand.java	org.soft2412.vsas.cli	75.24%	54.79%	105	48
ScrollUpdateSubcommand.java	org.soft2412.vsas.cli	79.12%	76.47%	91	29
UploadCommand.java	org.soft2412.vsas.cli	82.69%	66.67%	52	23
ScrollCommand.java	org.soft2412.vsas.cli	100.00%	100.00%	16	6
ProfileUpdateCommand.java	org.soft2412.vsas.cli	68.97%	72.41%	87	38
ScrollDeleteSubcommand.java	org.soft2412.vsas.cli	82.35%	74.07%	51	16
CommandDispatcher.java	org.soft2412.vsas.cli	65.00%	46.67%	20	14

10

entries per pageShowing 1 to 10 of 24 entries

123

Jenkins

A3-T28-003

main

#54

Pipeline Steps

Status

Changes

Console Output

Full Build Information

Delete build 'set'

Timings

Git Build Data

Tests

Coverage Report

JaCoCo HTML

Set Fingerprints

Open Blue Ocean

Pipeline Overview

Restart from Stage

Replay

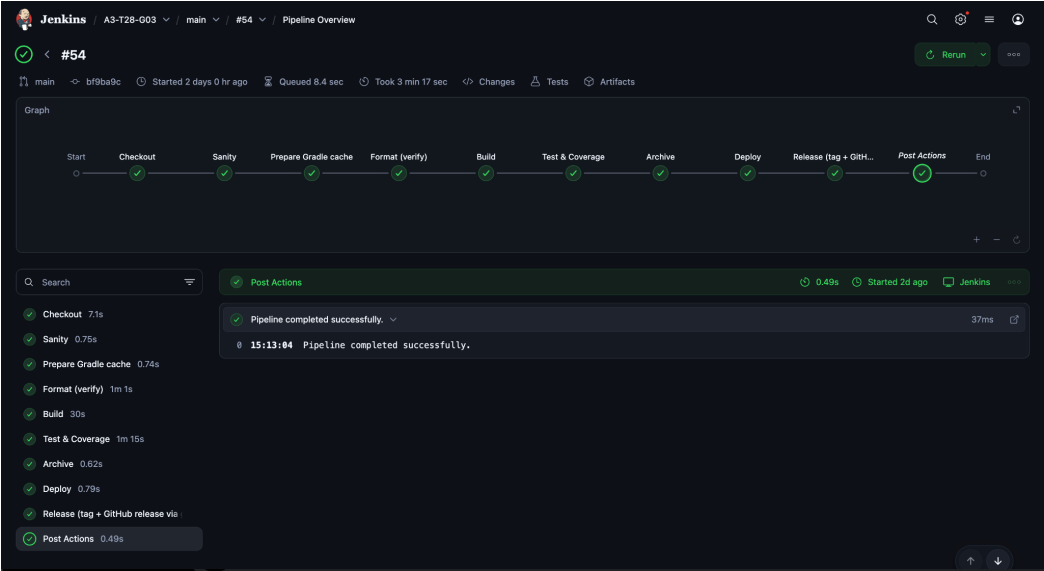
Pipeline Steps

Workspaces

Previous Build

Step	Arguments	Status
Start of Pipeline - (3 min 16 sec in block)		
node - (3 min 16 sec in block)		
node block - (3 min 16 sec in block)		
withEnv - (3 min 16 sec in block)	GITHUB_HOST, GRADLE_USER_HOME, GITHUB_REPO, GITHUB_OWNER, RELEASE_TEST_MODE	
withEnv block - (3 min 16 sec in block)		
timestamps - (3 min 16 sec in block)		
timestamps block - (3 min 16 sec in block)		
stage - (7.4 sec in block)	Checkout	
stage block (Checkout) - (7.1 sec in block)		
echo - (21 ms in self)	Checking out branch: main	
delete2 - (4.6 sec in self)		
checkout - (1.9 sec in self)		
stage - (1 sec in block)	Sentry	
stage block (Sentry) - (0.75 sec in block)		
sh - (0.41 sec in self)	echo "" PWD "" && pwd && echo "" Host listing "" && ls -la && echo "" Workspace listing "" && ls -la && gradlew gradlewProperties && true && echo "" GRADLE_USER_HOME "" && echo "" GRADLE_USER_HOME ""	
stage - (1 sec in block)	Prepare Gradle cache	
stage block (Prepare Gradle cache) - (0.74 sec in block)		
sh - (0.41 sec in self)	set -e mkdir -p "\$GRADLE_USER_HOME" chmod -R uwx "\$GRADLE_USER_HOME" & true	
stage - (1 min 2 sec in block)	Format (verify)	
stage block (Format (verify)) - (1 min 1 sec in block)		
echo - (31 ms in self)	Running google-java-format verify...	
script - (1 min 1 sec in block)		
script block - (1 min 1 sec in block)		
sh - (1 min 0 sec in self)	./gradlew --no-daemon -g "\$GRADLE_USER_HOME" verifyGoogleJavaFormat	
echo - (21 ms in self)	: Code format check passed.	
stage - (37 sec in block)	Build	
stage block (Build) - (30 sec in block)		
echo - (37 ms in self)	Building Gradle project...	
sh - (30 sec in self)	set -e chmod +x gradlew && true && build regular (src and the folder for ./gradlew --no-daemon -g "\$GRADLE_USER_HOME" clean assemble shadowJar	
stage - (1 min 16 sec in block)	Test & Coverage	
stage block (Test & Coverage) - (1 min 16 sec in block)		
echo - (32 ms in self)	Running unit tests and generating JaCoCo XML HTML reports...	
sh - (28 sec in self)	./gradlew --no-daemon -g "\$GRADLE_USER_HOME" test jacocoTestReport	
java - (13 sec in self)		
script - (16 sec in block)		
script block - (16 sec in block)		
recordCoverage - (14 sec in self)	(booker@parvus:ACOCO) pattern="" buildReportLocation="" jacocoTestReport.xml], sourceCodeLocation=CHECKOUT_BUILD, sourceDirectories=[path=app/src/main/java], [path=app/src/test/java]]	
publishHTML - (2.1 sec in self)		
archiveArtifacts - (16 sec in self)		
stage - (0.95 sec in block)	Archive	
stage block (Archive) - (0.82 sec in block)		
archiveArtifacts - (0.27 sec in self)		
stage - (11 sec in block)	Deploy	
stage block (Deploy) - (0.79 sec in block)		
echo - (32 ms in self)	Deploying application...	
sh - (0.41 sec in self)	echo "Deploy step placeholder. Add real deployment here if needed."	
stage - (15 sec in block)	Release (tag + GitHub release via gh)	
stage block (Release (tag + GitHub release via gh)) - (13 sec in block)		
withEnv - (12 sec in block)	GITHUB_HOST	
withEnv block - (12 sec in block)		
withCredentials - (1.2 sec in block)		
withCredentials block - (1.1 sec in block)		
sh - (11 sec in self)		
archiveArtifacts - (0.17 sec in self)		
stage - (0.76 sec in block)	Declarative: Post Actions	
stage block (Declarative: Post Actions) - (0.49 sec in block)		
echo - (37 ms in self)	Pipeline completed successfully.	

Jenkins 中文社区 Jenkins 2.528.1



Jenkins / A3-T28-G03 / main / #54 / Allocate node : Start / Workspace / Workspace

Search Status

Console Output

Workspace

Open Blue Ocean

Workspace

/

- .git
- .gradle
- .gradle-cache
- app
- coverage
- gradle
- .gitattributes Oct 24, 2025, 4:09:56 AM 214 B
- .gitignore Oct 24, 2025, 4:09:56 AM 815 B
- CHANGELOG.txt Oct 24, 2025, 4:13:01 AM 191 B
- coverage-sources.zip Oct 24, 2025, 4:12:29 AM 66.99 KiB
- gradle.properties Oct 24, 2025, 4:09:56 AM 25 B
- gradlew Oct 24, 2025, 4:09:56 AM 8.49 KiB
- gradlew.bat Oct 24, 2025, 4:09:56 AM 2.85 KiB
- Jenkinsfile Oct 24, 2025, 4:09:56 AM 8.75 KiB
- LICENSE Oct 24, 2025, 4:09:56 AM 1.06 KiB
- next-version.txt Oct 24, 2025, 4:13:00 AM 7 B
- README.md Oct 24, 2025, 4:09:56 AM 342 B
- settings.gradle Oct 24, 2025, 4:09:56 AM 517 B
- VERSION_MAJOR Oct 24, 2025, 4:09:56 AM 1 B

(all files in zip)

Timings

	Primary task	Including subtasks
In queue	Waiting	8.4 sec
	Blocked	0 ms
	Buildable	10 ms
	Total	8.4 sec
Building	3 min 17 sec	3 min 16 sec
Scheduled to completion		3 min 25 sec
Number of subtasks		1
Average executor utilization		1.0



Jenkins / A3-T28-G03 / main / #54 / Console Output

Timestamps

[View as plain text](#)

☒ System clock time

☒ Use browser timezone

☐ Elapsed time

☐ None

```
15:12:56 "active_lock_reason": null
15:12:56 }
15:12:56 ]
15:12:56 s/.*"head":{["~"]*"ref":"\[["~"]*\].*/\1/p
15:12:56
15:12:56 + SRC_BRANCH=
15:12:56 + [ -n ]
15:12:56 + sleep 4
15:13:00 + git log -1 --pretty=%s
15:13:00 + MERGE_SUBJ=Merge pull request #86 from SOFT2412-COMP9412-2025s2/chore/remove-legacy-list-output
15:13:00 + + printf %sgrep -Eiq
15:13:00 ^((feat|feature)/|)-
15:13:00 + + printf %sgrep -Eiq Merge pull request #86 from SOFT2412-COMP9412-2025s2/chore/remove-legacy-list-output
15:13:00 ^Merge pull request #[0-9]+ from \[^ ]+((feat|feature)/|)-
15:13:00 + + printf %sgrep -Eiq Merge pull request #86 from SOFT2412-COMP9412-2025s2/chore/remove-legacy-list-output
15:13:00 ^feat(|(|/|/)-)
15:13:00 + MINOR=8
15:13:00 + PATCH=2
15:13:00 + NEXT_VERSION=V2.8.2
15:13:00 + echo V2.8.2+
15:13:00 tee next-version.txt
15:13:00 V2.8.2
15:13:00 + git rev-parse -q --verify refs/tags/V2.8.2
15:13:00 + git tag -a V2.8.2 -m Release V2.8.2 (automated by Jenkins)
15:13:00 + git push origin V2.8.2
15:13:01 To https://github.sydney.edu.au/SOFT2412-COMP9412-2025s2/A3-T28-G03.git
15:13:01 * [new tag]
15:13:01 V2.8.2 -> V2.8.2
15:13:01 + [ -n V2.8.1..HEAD ]
15:13:01 + echo Changes since V2.8.1:
15:13:01 + echo
15:13:01 + git log --no-merges --pretty=%h %s (%an) V2.8.1..HEAD
15:13:01 + export GH_TOKEN=****
15:13:01 + export GH_HOST=github.sydney.edu.au
15:13:01 + gh release view V2.8.2 -R SOFT2412-COMP9412-2025s2/A3-T28-G03
15:13:01 + gh release create V2.8.2 -R SOFT2412-COMP9412-2025s2/A3-T28-G03 --target bf9ba9cb2a85fef8bc232a81dce6823fc74599c6 --title V2.8.2 --notes-file CHANGELOG.txt
15:13:02 https://github.sydney.edu.au/SOFT2412-COMP9412-2025s2/A3-T28-G03/releases/tag/V2.8.2
15:13:02 + echo Release V2.8.2 created via gh.
15:13:02 Release V2.8.2 created via gh.
15:13:02 + JAR_GLOB=app/build/libs/*-all.jar
15:13:02 + ls app/build/libs/app-all.jar
15:13:02 + echo Uploading fat jar(s):
15:13:02 Uploading fat jar(s):
15:13:02 + ls -lh app/build/libs/app-all.jar
15:13:02 -rw-r--r-- 1 jenkins jenkins 3.2M Oct 24 04:11 app/build/libs/app-all.jar
15:13:02 + gh release upload V2.8.2 app/build/libs/app-all.jar -R SOFT2412-COMP9412-2025s2/A3-T28-G03 --clobber
[Pipeline] archiveArtifacts
15:13:03 Archiving artifacts
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions) (hide)
[Pipeline] echo
15:13:04 Pipeline completed successfully.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // timestamps
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline

GitHub has been notified of this commit's build result

Finished: SUCCESS
```


3.3 Code Quality

3.3.1 Tooling and Linting

No material changes from last week. We continue to standardise formatting with google-java-format (Gradle plugin). Developers run `./gradlew googleJavaFormat` locally; Jenkins runs `verifyGoogleJavaFormat` on every build. Tests run with Gradle + JUnit 5, and JaCoCo generates coverage on each CI run (`jacocoTestReport`). GitHub still surfaces a green check on commits/PRs when CI passes.

3.3.2 Tests and Coverage

Unchanged in approach. We keep unit tests across CLI commands, repositories, the session service, and password hashing, and we publish JaCoCo reports on every build. The coverage workflow and reporting locations remain the same as last week; this sprint reuses that baseline without process changes.

3.3.3 Code Reviews

Process largely follows last week's plan, with one small refinement aligned to the rules noted elsewhere in this report: reviewer approval is now enforced by the protected-branch rules, in

addition to the existing “CI must be green” gate. Otherwise, our review practice and PR conventions are unchanged.

3.3.4 Coding Standards

No changes. We continue to rely on Google Java Style via google-java-format, keep public contracts stable, and make small, focused commits that reference task IDs with conventional prefixes

```
kscii@mac1 A3-T28-G03 % tree app/src
app/src
├── main
│   ├── java
│   │   ├── org
│   │   │   ├── soft2412
│   │   │   │   ├── vsas
│   │   │   │   │   ├── App.java
│   │   │   │   │   ├── cli
│   │   │   │   │   │   ├── Command.java
│   │   │   │   │   │   ├── CommandDispatcher.java
│   │   │   │   │   │   ├── DownloadCommand.java
│   │   │   │   │   │   ├── ListCommand.java
│   │   │   │   │   │   ├── LoginCommand.java
│   │   │   │   │   │   ├── LogoutCommand.java
│   │   │   │   │   │   ├── PasswordPrompt.java
│   │   │   │   │   │   ├── PreviewCommand.java
│   │   │   │   │   │   ├── ProfileUpdateCommand.java
│   │   │   │   │   │   ├── RegisterCommand.java
│   │   │   │   │   │   ├── ScrollCommand.java
│   │   │   │   │   │   ├── ScrollDeleteSubcommand.java
│   │   │   │   │   │   ├── ScrollUpdateSubcommand.java
│   │   │   │   │   │   ├── UploadCommand.java
│   │   │   │   │   │   ├── WhoAmICommand.java
│   │   │   │   │   ├── model
│   │   │   │   │   │   ├── Scroll.java
│   │   │   │   │   │   ├── User.java
│   │   │   │   │   ├── repo
│   │   │   │   │   │   ├── FileScrollRepository.java
│   │   │   │   │   │   ├── FileUserRepository.java
│   │   │   │   │   │   ├── ScrollRepository.java
│   │   │   │   │   │   ├── UserRepository.java
│   │   │   │   │   ├── security
│   │   │   │   │   │   ├── PasswordHasher.java
│   │   │   │   │   │   ├── SessionService.java
│   │   │   │   │   ├── service
│   │   │   │   │   │   ├── SessionService.java
│   │   │   │   │   ├── test
│   │   │   │   │   │   ├── java
│   │   │   │   │   │   │   ├── org
│   │   │   │   │   │   │   │   ├── soft2412
│   │   │   │   │   │   │   │   │   ├── vsas
│   │   │   │   │   │   │   │   │   │   ├── AppInteractiveTest.java
│   │   │   │   │   │   │   │   │   │   ├── AppTest.java
│   │   │   │   │   │   │   │   │   │   ├── cli
│   │   │   │   │   │   │   │   │   │   │   ├── AuthHashingIntegrationTest.java
│   │   │   │   │   │   │   │   │   │   │   ├── CommandDispatcherTest.java
│   │   │   │   │   │   │   │   │   │   │   ├── DownloadCommandTest.java
│   │   │   │   │   │   │   │   │   │   │   ├── LoginCommandInteractiveTest.java
│   │   │   │   │   │   │   │   │   │   │   ├── LoginCommandTest.java
│   │   │   │   │   │   │   │   │   │   │   ├── LogoutCommandTest.java
│   │   │   │   │   │   │   │   │   │   │   ├── PasswordPromptTest.java
│   │   │   │   │   │   │   │   │   │   │   ├── PreviewCommandTest.java
│   │   │   │   │   │   │   │   │   │   │   ├── ProfileUpdateCommandTest.java
│   │   │   │   │   │   │   │   │   │   │   ├── RegisterCommandInteractiveTest.java
│   │   │   │   │   │   │   │   │   │   │   ├── RegisterCommandTest.java
│   │   │   │   │   │   │   │   │   │   │   ├── RegisterCommandUniqueKeyIdTest.java
│   │   │   │   │   │   │   │   │   │   │   ├── ScrollCommandTest.java
│   │   │   │   │   │   │   │   │   │   │   ├── ScrollDeleteSubcommandTest.java
│   │   │   │   │   │   │   │   │   │   │   ├── ScrollUpdateSubcommandTest.java
│   │   │   │   │   │   │   │   │   │   │   ├── FileScrollRepositoryTest.java
│   │   │   │   │   │   │   │   │   │   │   ├── ListCommandFiltersAndFormattingTest.java
│   │   │   │   │   │   │   │   │   │   │   ├── ListCommandTest.java
│   │   │   │   │   │   │   │   │   │   │   ├── model
│   │   │   │   │   │   │   │   │   │   │   │   ├── UserTest.java
│   │   │   │   │   │   │   │   │   │   │   ├── repo
│   │   │   │   │   │   │   │   │   │   │   │   ├── FileScrollRepositoryEdgeTest.java
│   │   │   │   │   │   │   │   │   │   │   │   ├── FileUserRepositoryDuplicateTest.java
│   │   │   │   │   │   │   │   │   │   │   │   ├── FileUserRepositoryTest.java
│   │   │   │   │   │   │   │   │   │   │   ├── security
│   │   │   │   │   │   │   │   │   │   │   │   ├── PasswordHasherTest.java
│   │   │   │   │   │   │   │   │   │   │   │   ├── SessionServiceTest.java
│   │   │   │   │   │   │   │   │   │   │   │   ├── UploadCommandTest.java
│   │   │   │   │   │   │   │   │   │   │   │   ├── WhoAmICommandTest.java
│   │   │   │   │   │   │   │   │   │   │   │   │   ├── 28 directories, 52 files
└── test
```

4. Application Development

4.1 Collaborative Coding

In the Sprint 2, the team continued to adopt the feature-branch workflow to maintain the orderliness and reliability of the development process. Each member developed the assigned user stories on independent branches and followed a unified naming convention: new features were named feat/<feature-name>, and bug fixes were named fix/<issue-name>.

Throughout the entire sprint, the team regularly synchronizes the latest updates of the main branch to avoid merge conflicts. When conflicts arise, members resolve them manually and document the handling process in the Pull Request description. All feature branches must pass peer review and continuous integration (CI) checks before being merged to ensure the correctness of the functionality and the quality of the code.

The team also strictly adheres to the Google Java coding standards and Conventional Commits commit format to maintain a consistent code style and commit records. The correctness and

coverage of the code are verified by running automated tests (./gradlew test jacocoTestReport), and the average line coverage of newly implemented command functions exceeds 80%.

fix/login-single-prompt-and-usage	2 days ago	3 / 3	4 / 0	#85	...
feat/us-a3-update-profile	2 days ago	3 / 3	7 / 0	#84	...
feat/us-e1-track-scroll-usage	2 days ago	3 / 3	17 / 0	#82	...
feat/us-d4-preview-scroll	2 days ago	3 / 3	20 / 0	#81	...
feat/d3-search-filters	3 days ago	3 / 3	38 / 0	#80	...
fix/release-feat-detection-anchor	3 days ago	3 / 3	42 / 0	#76	...
feat/g1-secure-password-prompt	3 days ago	3 / 3	39 / 0	#79	...
feat/scroll-update	3 days ago	3 / 3	45 / 0	#78	...
fix/release-feat-detection	3 days ago	3 / 3	45 / 0	#75	...

app

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
org.soft2412.vsas.cli		83%		65%	114	210	100	488	17	54	0	12
org.soft2412.vsas.repo		90%		62%	51	94	34	193	0	24	0	2
org.soft2412.vsas.service		78%		50%	13	23	13	55	0	9	0	1
org.soft2412.vsas		77%		65%	12	26	15	62	1	6	0	1
org.soft2412.vsas.security		91%		80%	4	17	4	46	0	7	0	1
org.soft2412.vsas.model		87%		100%	1	19	1	35	1	18	0	2
Total	560 of 3,838	85%	185 of 524	64%	195	389	167	879	19	118	0	19

4.2 Individual and Group Contributions

Xuanjian Fang:

Delivered US-G1: Secure password input (masked prompt).

Delivered US-D3: Search & filters.

Owned CI/CD integration (pipeline setup & maintenance), test automation (gradle + Jacoco in CI), and automatic versioning/tagging during releases.

Zhiwei Cheng:

Delivered US-E1: Track upload/download counts.

Delivered US-D2: Download from list.

Co-delivered US-D4: Preview scrolls with Boning Zhao.

Boning Zhao:

Delivered US-A3: Update profile.

Co-delivered US-D4: Preview scrolls with Zhiwei Cheng.

Authored User Stories.

Xuanhao Yu:

Delivered US-C3: Remove own scroll.

Delivered US-C2: Edit own scroll.

☐	US-G1 Secure password input (masked prompt) #71 by bzha8676 was closed yesterday 4 tasks done	2
☐	US-E1 Track upload/download counts #70 by bzha8676 was closed yesterday 4 tasks done	1
☐	US-D3 Search & filters #69 by bzha8676 was closed yesterday 4 tasks done	1
☐	US-D2 Download from list #68 by bzha8676 was closed yesterday 5 tasks done	1
☐	US-C3 Remove own scroll #67 by bzha8676 was closed yesterday 4 tasks done	1
☐	US-C2 Edit own scroll #66 by bzha8676 was closed yesterday 4 tasks done	1
☐	US-A3 Update profile #64 by bzha8676 was closed yesterday 5 tasks done	1
☐	US-D4 Preview scrolls #63 by bzha8676 was closed yesterday 4 tasks done	

V2.8.2

...

🕒 2 days ago

🔑 bf9ba9c

📦 zip

📦 tar.gz

📄 Notes

⬇️ Downloads

V2.8.1

...

🕒 2 days ago

🔑 353249e

📦 zip

📦 tar.gz

📄 Notes

⬇️ Downloads

V2.8.0

...

🕒 2 days ago

🔑 f292607

📦 zip

📦 tar.gz

📄 Notes

⬇️ Downloads

V2.7.0

...

🕒 2 days ago

🔑 872c4a5

📦 zip

📦 tar.gz

📄 Notes

⬇️ Downloads

V2.6.0

...

🕒 2 days ago

🔑 be1a5c1

📦 zip

📦 tar.gz

📄 Notes

⬇️ Downloads

V2.5.0

...

🕒 3 days ago

🔑 7898e5a

📦 zip

📦 tar.gz

📄 Notes

⬇️ Downloads

V2.4.0

...

🕒 3 days ago

🔑 7898e5a

📦 zip

📦 tar.gz

📄 Notes

⬇️ Downloads

V2.3.1

...

🕒 3 days ago

🔑 c9ea8bb

📦 zip

📦 tar.gz

📄 Notes

⬇️ Downloads

V2.3.0

...

🕒 3 days ago

🔑 c2f23fc

📦 zip

📦 tar.gz

📄 Notes

⬇️ Downloads

2 days ago

xfan0282

V2.8.2

bf9ba9c

Compare

V2.8.2

Latest

Changes since V2.8.1:

- [20ffe69](#) test(list): update tests to verify fixed-width output format (bzha8676)
- [f851290](#) refactor(list): remove legacy output for backwards compatibility (bzha8676)

▼ Assets 3

app-all.jar	3.17 MB	2 days ago
Source code (zip)		2 days ago
Source code (tar.gz)		2 days ago

😊

2 days ago

xfan0282

V2.8.1

353249e

Compare

V2.8.1

Changes since V2.8.0:

- [82e5edd](#) test(login): add tests for single password prompt and fast-fail usage when username missing (xfan0282)
- [779dd79](#) fix(login): prompt for password once; require username before prompting and show usage (xfan0282)

► Assets 3

😊

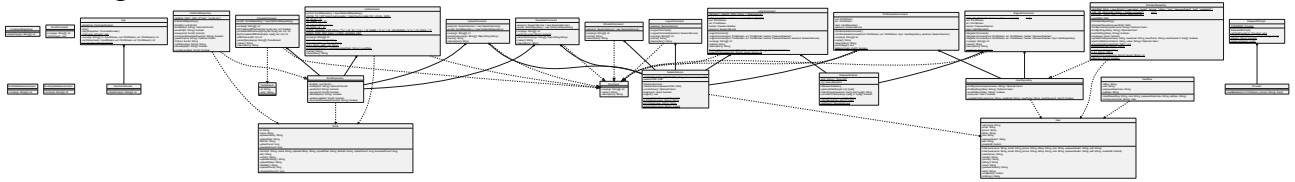
4.3 CI (Continuous Integration) Tools and Structure Management

In Sprint 2, the team continued to use the Jenkins continuous integration (CI) workflow that Xuanjian Fang set up during Sprint 1, to ensure that each code update could automatically perform building, testing and reporting, thereby maintaining the stability of the project. This pipeline adopts the multi-branch structure defined by Jenkinsfile. It is automatically triggered each time a Pull Request is made or a commit is made on the main branch, and the execution sequence is as follows: Checkout → Build → Test → Coverage Report → Archive. Gradle is responsible for automating build and testing tasks, while the Jacoco plugin automatically generates coverage data after each test and sets a minimum 75% coverage threshold as the standard for a successful build. This mechanism promotes test-driven development (TDD) and prevents unverified code from being merged. All CI results (including Jacoco reports) are archived in HTML format for the team to review and track quality on a regular basis.

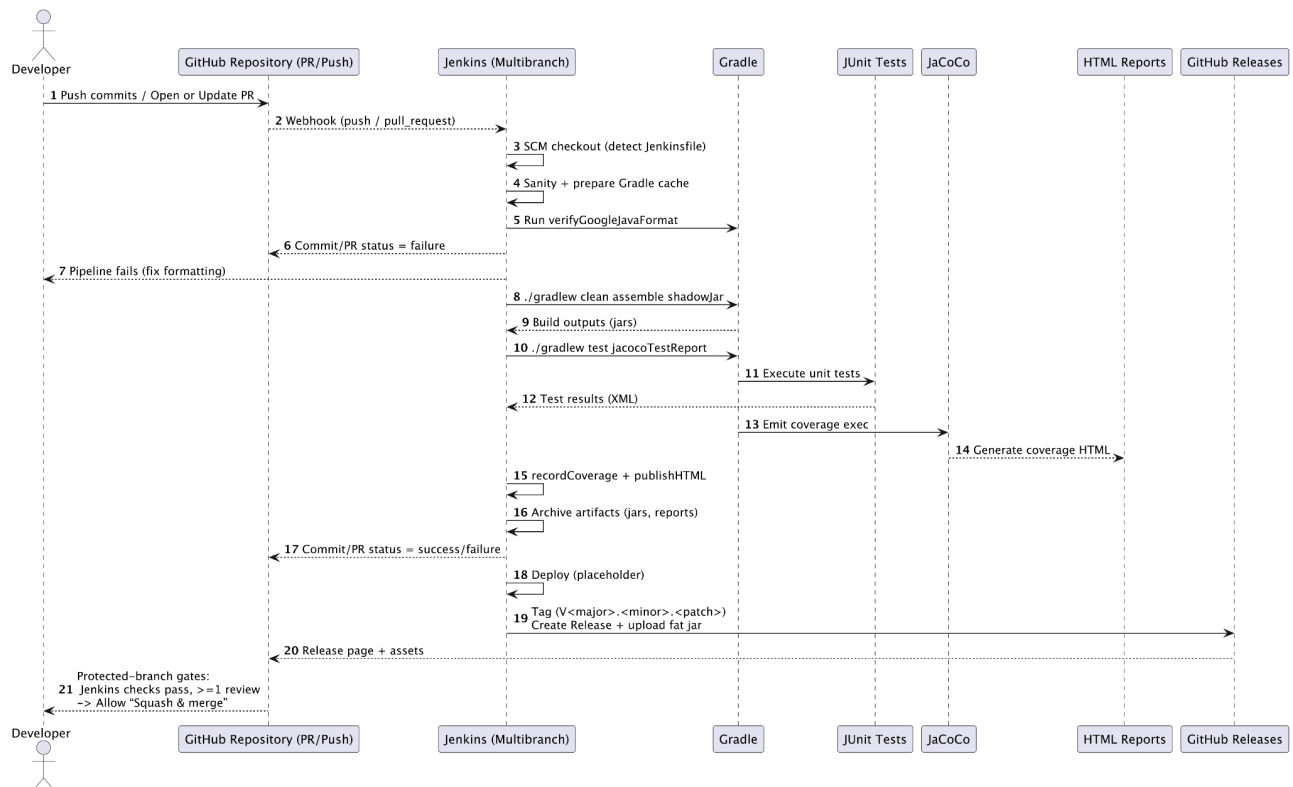
In terms of structure, the system is composed of multiple Gradle sub-modules - CLI, Repository, Service and Security - which support independent development and parallel testing. This modular

design not only reduces merge conflicts but also simplifies CI configuration, enabling parallel task execution and unified coverage summary.

class diagram:



Sequence Diagram:



4.4 Functional and Non-functional Aspects

In Sprint 2, the system has significantly enhanced the integrity and user experience of the scroll management function. The application now supports previewing, editing, deleting, downloading and searching of scrolls, as well as updating user profiles.

Functional improvements

Scroll Preview (US-D4): Users can view detailed information about the scroll, including the scroll ID, name, upload time, and the first line of the scroll content. They can quickly understand the content without having to open the file.

Scroll Update (US-C2): Allows users to modify their own scrolls, including updating metadata or replacing file contents, and performs ownership verification to prevent unauthorised modifications.

Scroll deletion (US-C3): Allows users to permanently delete their scrolls. The system provides a [y/n] confirmation prompt before execution to prevent accidental deletion.

Scroll download (US-D2): Users can choose to download to the current folder (the default option) or specify a different folder, enhancing flexibility and file management efficiency.

User Profile Update (US-A3): Allows users to modify their personal information (email address, password, mobile phone number).

Scroll Search (US-D3): A new condition-based search function has been added, allowing you to retrieve the target scroll based on scroll ID, name, or upload time range.

Non-functional improvements

Reliability: Through the Jenkins continuous integration pipeline, automatic building and testing are carried out to ensure that each submission is stable and usable.

Maintainability: The modular structure of Gradle (CLI, Repository, Service, Security) reduces coupling and facilitates independent testing and debugging.

Security: Authorization verification, password mask input, and login session checks ensure data security.

```
vsas> list
id      name      uploader  uploadDate
1       ReadMe     1         2025-10-24T04:25:30.
2       gitignore  1         2025-10-26T04:23:40.
vsas> whoami
test1 (role=USER)
vsas> scroll update --id 1 --name "readme" --file "C:\Users\32234\code\Java\2412-2025\A3-T28-G03\README.md"
Replace existing file? [y/n] y
Updated: 1
vsas> list
id      name      uploader  uploadDate
1       readme    1         2025-10-24T04:25:30.
2       gitignore 1         2025-10-26T04:23:40.
vsas> preview --id 1
id: 1
name: readme
uploader: 1
uploadDate: 2025-10-24T04:25:30.101535100Z
size: 361 bytes
text: # CC28-Team03-A2A3 - VSAS    ## Run    ./gradlew run    ## Test

vsas> list --uploader-id 1
id      name      uploader  uploadDate
1       readme    1         2025-10-24T04:25:30.
2       gitignore 1         2025-10-26T04:23:40.
vsas> list --scroll-id 1
id      name      uploader  uploadDate
1       readme    1         2025-10-24T04:25:30.
vsas> list --name readme
id      name      uploader  uploadDate
1       readme    1         2025-10-24T04:25:30.

vsas> profile update --email 123@321.com
Updated: test1.
vsas> download --id 1
Use current directory? [y/N] y
C:\Users\32234\code\Java\2412-2025\A3-T28-G03\app\1. bin
vsas> download --id 1 --out C:\Users\32234\code\Java\2412-2025
C:\Users\32234\code\Java\2412-2025\1. bin
vsas> list
id      name      uploader  uploadDate
1       readme    1         2025-10-24T04:25:30.
2       gitignore 1         2025-10-26T04:23:40.
vsas> scroll delete --id 1
Delete scroll 1? This cannot be undone. [y/n] y
Deleted: 1
vsas> list
id      name      uploader  uploadDate
2       gitignore 1         2025-10-26T04:23:40.
```

5. Group Contribution

5.1 Team Organisation and Roles

The team followed the Scrum methodology to maintain a self-organised and iterative development process.

Roles were clearly defined in accordance with the Scrum structure:

Product Owner – Boning Zhao

Responsible for defining user requirements, prioritising backlog items, and ensuring that implemented features aligned with stakeholder expectations.

In addition to managing the product backlog, he contributed to the system's architecture by creating the initial Gradle-based scaffold and standardising the project's directory hierarchy.

Scrum Master – Xuejian Fang

Oversaw sprint planning, progress tracking, and facilitated team communication to ensure adherence to Scrum principles.

From a technical perspective, he implemented the CI/CD infrastructure, configuring the Jenkins multibranch pipeline and integrating Jacoco for automated coverage validation.

Core Team Members – Other Developers

Implemented assigned functional modules according to their respective Epics and User Stories, including user registration, authentication, and scroll management features. Each member maintained testing accountability for their code.

This role allocation balanced project management responsibilities and technical development, enabling the team to maintain both delivery speed and software quality throughout the sprint.

5.2 Communication and Collaboration Tools

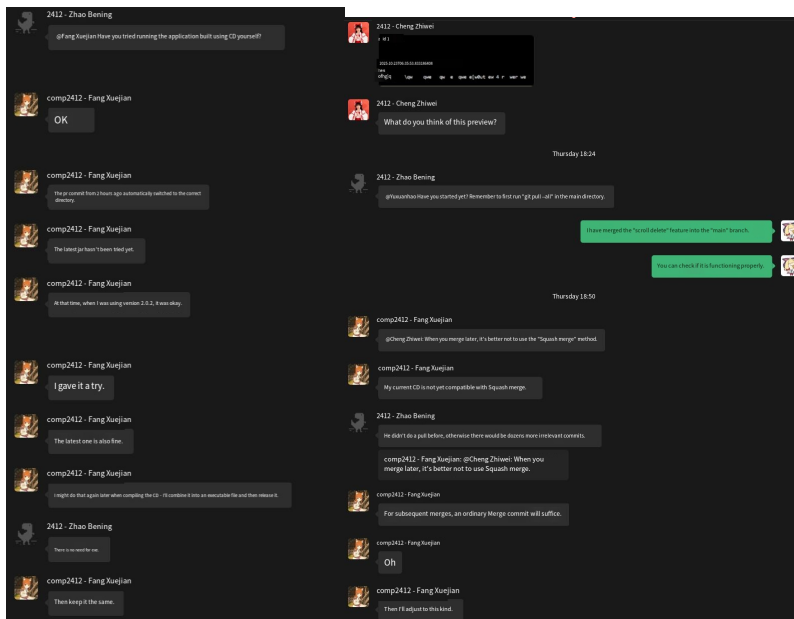
In sprint 2, our main mode of communication was offline meetings, supplemented by WeChat groups for quick online coordination.

Most discussions and decisions were made during face-to-face meetings. This approach made communication more efficient, as issues could be resolved on the spot. During offline meetings, we would review the progress of user stories (US), discuss details of function implementation, assign development tasks, and resolve code merge conflicts on the spot. Face-to-face

communication also further enhanced the cooperation awareness and understanding among team members.

Apart from offline meetings, we use WeChat groups for asynchronous communication, which is used for progress updates, emergency issue discussions, and quick confirmations. This way, even when team members are not in the same location or working at different times, they can stay in touch. The WeChat groups are also used to share screenshots, submit record links, and functional demonstration videos, making it convenient for team members to promptly understand each other's work results.

In terms of version control and task collaboration, we still use GitHub as our main development platform. It supports branch collaboration, issue tracking, and Pull Request review, ensuring that all contributions from team members can be recorded, verified, and safely merged.



5.3 Group Discussion

In Sprint 2, our team held several offline meetings to discuss various matters, including design decisions, progress reports, and task coordination. We ensured that each User Story was implemented in accordance with the unified standards and was consistent with the overall project goals.

At the beginning of the sprint, we reviewed the product backlog and selected the development goals for Sprint 2 based on dependencies and priorities, including scroll preview, updates, deletion, user profile modification, and search functionality. During the meeting, the team members proposed different technical solutions, compared various design ideas, and clarified the interfaces between the modules.

In addition, we also resolved merge conflicts and testing issues through group discussions. When two members made overlapping modifications in the same class, the team would jointly view the two implementations in VS Code, evaluate their pros and cons, and decide on the merging strategy. This discussion reduced errors and made the team's decision more fair and reasonable.

After each major milestone is completed, we also hold periodic review meetings. Each member presents the functions they have completed (such as new commands or optimised interface interaction processes), and receives feedback from other members. Through this peer review mechanism, we can identify usability issues in advance.

5.4 Individual Contributions

Xuanjian Fang is responsible for implementing US-G1 and US-D3, and has undertaken the configuration work of continuous integration (CI/CD), automated testing, and automatic version number generation. He achieved automatic building and checking through GitHub Actions, ensuring that style checks and tests are automatically executed for every commit and Pull Request.

Zhiwei Cheng was responsible for developing US-E1 and US-D2. Additionally, he collaborated with Zhao Bonian to complete US-D4.

Boning Zhao was mainly responsible for implementing US-A3 and jointly completed US-D4 with Zheng Zhiwei. Additionally, he was also in charge of writing and refining the User Stories documentation.

Xuanhao Yu completed US-C2 (scroll update) and US-C3 (scroll deletion).

5.5 Additional Notes and Reflections

The Sprint 2 was a significant improvement for our team. Not only did it yield remarkable results in project progress, but it also made the team more mature in terms of collaboration and professional practice. Compared to the first sprint, this iteration has seen notable improvements in collaboration coordination, task allocation, and adherence to engineering standards.

One of the most profound reflections from this sprint is the importance of planning and communication. In the first sprint, due to unclear task ownership and insufficient communication, some function development was delayed. This time, through regular offline meetings and synchronous WeChat groups, we significantly improved this issue. Risks could be identified earlier, task allocation was more reasonable, thus making the development cycle smoother and the tension before the deadline was significantly reduced.

We have also made significant progress in testing and automation processes. Through continuous integration (CI) and automated test reports, we can monitor code quality and coverage in real time, and promptly identify potential issues. The average code coverage of the final system exceeded 80%, much higher than the level of the first sprint, and the stability of the project has significantly improved.

Looking forward to the next sprint, we plan to focus on usability optimization and document consistency. Although the command-line functionality is already quite complete, there is still room for improvement in input processing and feedback prompts. At the same time, we will ensure that the documentation is updated in sync with the development to enhance the maintainability of the system.

Overall, the second sprint was a successful iteration, fully demonstrating the team's growth in technical capabilities, problem-solving skills, and professional collaboration.